

Derivative Products and Their Pricing — Interest Rate (IR) Module

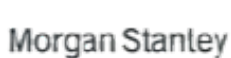
Top 15 IR products — Advanced Quant Guide

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8 8. Bermudan Swaption

Definition and Market Context

A Bermudan Swaption is an option to enter into a swap at multiple discrete exercise dates, typically on coupon payment dates. Unlike European (single expiry) or American (continuous exercise), Bermudan offers a middle ground—exercise only on specific dates. Example: 10-year swap callable after 1 year, exercisable annually. Bermudans are core to callable debt and MBS hedging. Pricing requires numerical methods (trees, Monte Carlo) as no closed-form exists. The exercise boundary depends on future volatility, rates, and the "value of waiting."

Intuition: The "Island Hopping" Ticket

A European swaption is a ticket to a single destination. A Bermudan swaption is a "hop-on, hop-off" pass.

- You have the right to enter the swap on *any* anniversary date (island) along the journey.
- Once you use the ticket (exercise), the option is gone.
- Because you have the flexibility to wait for the perfect moment (highest rate for payer, lowest for receiver), it costs more than a standard European option.

Quick Formula Sheet: Bermudan Swaption

- **Valuation:** Backward Induction (Tree or Monte Carlo).
- **Decision Rule:** Exercise if Intrinsic Value > Continuation Value.
- **Premium:** $\text{Value}_{\text{Bermudan}} \approx \text{Value}_{\text{Euro}} + \text{Flexibility Value}$.

Why Banks and Hedge Funds Use Bermudan Swaptions

- **Callable Debt Hedging:** Corporations issue 5-year bonds callable after 2 years. The call right is a Bermudan receiver swaption. Investment banks buy Bermudan swaptions to hedge the call feature.
- **Mortgage Servicing:** Mortgage prepayment option is Bermudan (homeowners can prepay on any payment date). Servicers buy Bermudan receiver swaptions to hedge prepayment risk as rates fall.
- **Relative Value:** Bermudan swaptions trade rich to European strips due to exercise flexibility. Hedge funds sell Bermudan, buy strip of Europeans, capturing the "option on option" premium.
- **Exotic Structuring:** Callable range accruals and inverse floaters embed Bermudan features. Dealers calibrate LSV (local stochastic vol) models to Bermudan prices to capture volatility dynamics.
- **Liquidity Premium:** Bermudan options incorporate liquidity value—ability to exit position on specific dates reduces funding risk for banks.

Detailed Cashflow Mechanics

Exercise at Time t_k : If exercised at date t_k , receive swap from t_k to T with strike K . The continuation value at t_k is:

$$C(t_k) = \max \left(\text{SwapPV}(t_k), \mathbb{E}^Q \left[\frac{C(t_{k+1})}{M(t_k, t_{k+1})} \right] \right)$$

where M is money market account. **Backward Induction:** Starting at maturity T :

$$V(T) = 0 \quad (\text{no exercise past maturity})$$

At exercise date t_k :

$$V(t_k) = \max \left(\underbrace{A(t_k)(S(t_k) - K)}_{\text{Exercise Value}}, \underbrace{\mathbb{E}^Q \left[\frac{V(t_{k+1})}{M(t_k, t_{k+1})} \right]}_{\text{Continuation Value}} \right)$$

12 12. Callable & Puttable Bond

Definition and Market Context

A **callable bond** gives the issuer the right to redeem before maturity at a predetermined call price (typically par). A **puttable bond** gives the investor the right to sell back to the issuer. These embedded options create Bermudan exercise features (exercisable on coupon dates). Callable bonds trade at higher yield than non-callable to compensate investors for call risk. Puttable bonds trade at lower yield due to investor's optionality. Pricing requires backward induction (trees) to capture optimal exercise decisions.

Intuition: The Refinance Option

A Callable Bond is a bond where the issuer (borrower) has a "Do-Over" button.

- If rates drop significantly, the issuer hits the button (Calls the bond), pays you back par, and issues new debt at the lower rate.
- Because they have this advantage, they must pay you a higher yield today.
- **Risk:** As an investor, your upside is capped (price won't go much above par), but your downside is full (if rates rise, price crashes). This is **Negative Convexity**.

Quick Formula Sheet: Callable Bond

- **Valuation:** $P_{callable} = P_{straight} - P_{call_option}$
- **OAS:** The "true" spread earned after stripping out the option cost.
- **Behavior:** Price hits a ceiling as yields fall.

Why Banks and Hedge Funds Use Callable/Putable Bonds

- **Refinancing Flexibility:** Corporations issue callable bonds to refinance if rates fall, avoiding being locked into high coupon debt. The call option has economic value worth 20-50bp in yield.
- **Put Protection:** Investors buy puttable bonds to protect against rate rises. If rates spike, they can put the bond back at par and reinvest at higher yields.
- **Convexity Arbitrage:** Callable bonds have negative convexity (issuer will call when rates fall). Dealers hedge by buying receiver swaptions, creating "option-adjusted spread" (OAS) trades.
- **Liability Management:** Banks issue callable subordinated debt to manage regulatory capital. If capital requirements change, they can call and reissue.
- **Structured Products:** Callable range accruals and step-up callable notes embed Bermudan features that require sophisticated modeling for hedging.

Detailed Cashflow Mechanics

Callable Bond: At each call date t_k , issuer decides:

$$\text{Call if } V_{\text{continuing}}(t_k) > \text{Call Price}$$

where $V_{\text{continuing}}$ is present value of remaining cashflows.

Puttable Bond: At each put date t_k , investor decides:

$$\text{Put if Put Price} > V_{\text{continuing}}(t_k)$$

Price Decomposition:

$$P_{\text{callable}} = P_{\text{straight}} - V_{\text{call option}}$$

$$P_{\text{puttable}} = P_{\text{straight}} + V_{\text{put option}}$$

```

11     """CMS spread option specifications"""
12     notional: float
13     expiry_date: datetime
14     payment_date: datetime
15     long_cms_tenor: int # e.g., 10 for 10Y CMS
16     short_cms_tenor: int # e.g., 2 for 2Y CMS
17     strike: float # Spread strike in bps (e.g., 0.5% = 0.005)
18     option_type: str = 'call' # 'call' or 'put' on spread
19
20 class CMSSpreadOption:
21     """Production-grade CMS spread option with correlation modeling"""
22
23     def __init__(self,
24                 specs: CMSSpreadOptionSpecs,
25                 discount_curve: DiscountCurve,
26                 long_smile: SwaptionSmile,
27                 short_smile: SwaptionSmile,
28                 correlation: float = 0.8):
29         self.specs = specs
30         self.curve = discount_curve
31         self.long_smile = long_smile
32         self.short_smile = short_smile
33         self.rho = correlation
34
35         # Calculate forwards and adjustments
36         self.long_fwd = self._cms_forward(specs.long_cms_tenor)
37         self.short_fwd = self._cms_forward(specs.short_cms_tenor)
38         self.long_ca = self._convexity_adjustment(specs.long_cms_tenor)
39         self.short_ca = self._convexity_adjustment(specs.short_cms_tenor)
40
41         # Expected CMS rates
42         self.long_rate = self.long_fwd + self.long_ca
43         self.short_rate = self.short_fwd + self.short_ca
44
45         # Spread forward
46         self.spread_fwd = self.long_rate - self.short_rate
47
48     def _cms_forward(self, tenor: int) -> float:
49         """Simplified CMS forward calculation"""
50         start = self.specs.expiry_date
51         end = start + timedelta(days=365 * tenor)
52         df_start = self.curve.df(start)
53         df_end = self.curve.df(end)
54
55         # Approximate annuity
56         years = np.arange(0.5, tenor + 0.5, 0.5)
57         annuity = sum(0.5 * self.curve.df(start + timedelta(days=int(365*y))) for y in
58                       years)
59
60         return (df_start - df_end) / annuity
61
62     def _convexity_adjustment(self, tenor: int) -> float:
63         """Convexity adjustment for CMS rate"""
64         expiry = (self.specs.expiry_date - self.curve.today).days / 365.0
65
66         # Use respective smile
67         smile = self.long_smile if tenor == self.specs.long_cms_tenor else self.
68             short_smile
69
70         # Simplified: CA = 0.5 * rho * T * duration

```